

Mission Antyodaya 2020 Cluster Analysis

What 6 Lakh Indian Villages Tell Us About Rural Development

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The Mission Antyodaya 2020 survey records rural development through a large number of variables covering public services, institutions, infrastructure, livelihoods, agriculture, health, water, sanitation, and connectivity. Read separately, these variables are difficult to compare and do not readily show how different aspects of a village interact. Our analysis reduces this complexity to a smaller set of indices while retaining the component features needed for diagnosis and further analysis.

Data Preparation

Of the 198 parameters available across 32 state and union-territory files, 121 were selected for analysis. They were transformed into 64 features using normalization, binary availability measures, inversion, ratios, ordinal mappings, and composite scores. All features were oriented so that a higher value represents more of the measured positive condition. Examples include piped-water coverage, hours of domestic electricity, child nutrition and development, land utilization, agricultural risk support, and access to markets.

The 64 features were organized into 21 categories. Features were classified as Low, Medium, and High where sufficient variation existed; binary indicators used Low and High. Within each category, these classes were scored (Low = 0, Medium = 0.5, High = 1) and combined with equal weight. MiniBatch K-means was then used to classify the resulting category index. Each category is represented by a normalised value and a final class.

How to Read a Category

Begin with a `*_cat_cluster` field. It gives the final Low, Medium, or High class for one of the 21 categories. Next read the matching `*_cat_value`, then examine the normalised `*_feat_value` fields that contribute to it. These feature values are comparative measures on a common 0–1 scale. Raw survey fields show the reported entries used to construct the features. They are useful for tracing a result, but should not be used to rank villages because they have different units and have not been normalised.

Most categories use Low, Medium, and High. Four categories contain only Low and High:

- Livelihoods: Cottage and Traditional Industry;
- Livelihoods: Forest Resources;
- Livelihoods: Common Resources; and
- Livelihoods: Alternative Farming.

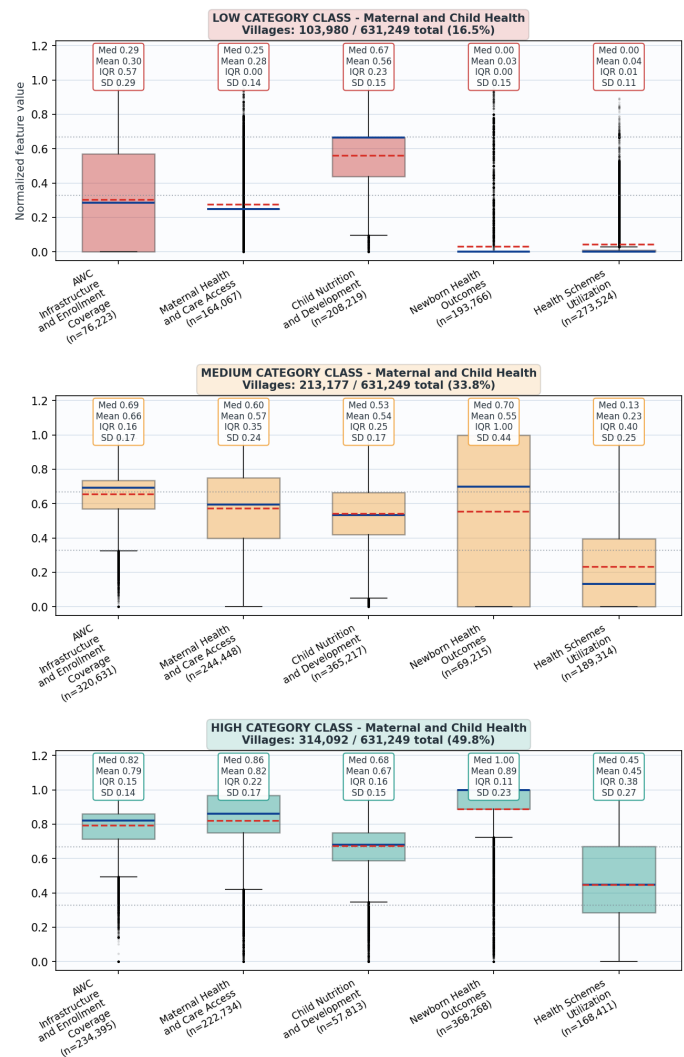
When these four categories are used as filters, a Medium class should not be expected.

Reading the Boxplot

Figure 1 shows how to open a category and examine its components. Each panel contains villages in one final Maternal and Child Health category class: Low, Medium, or High. Within a panel, each box shows one of the five normalised feature values. The blue line is the median, the red dashed line is the mean, the box covers the middle half of observed values, and the whiskers and points show the wider spread.

The panels are separate groups of villages, not a time sequence. Read one panel at a time and compare the five components within it. In the High panel, newborn health outcomes and maternal health and care access sit relatively high, while health-scheme utilisation sits lower and is more widely spread. The final category class summarises the five components together; the component values show which parts of that summary are stronger or weaker.

Maternal and Child Health



Blue line = median | Red dashed line = mean | Box = Q1-Q3 | Whiskers = 1.5x IQR | n = villages in that feature class

Figure 1: Maternal and Child Health feature distributions across the three category classes.

The boxplot therefore answers a simple question: within each final category class, which components are consistently strong, which are weaker, and which vary widely among villages?

Using Category Patterns

Only the 21 category classes should be used to define patterns or filter villages. The category value and normalised feature values can then explain why a selected village falls within that pattern. For example:

Water-security pressure.

Pattern: High Livelihoods: Employment + Medium/High Agriculture: Land Cultivation + Low Agriculture: Irrigation and Watershed.

This combination points to places where livelihood dependence and cultivation are substantial but the measured water-security system is weak.

Agriculture without protective systems.

Pattern: Medium/High Agriculture: Land Cultivation + Low Agriculture: Support Services + Low Agricultural Markets + Low Financial Inclusion.

The filter brings together cultivation with weak support, finance, storage, and market access.

Institutions without economic conversion.

Pattern: High Institutionalization + Low Financial Inclusion + Low Agricultural Markets.

A stronger institutional base is present, but links to finance and markets remain weak.

Connectivity without local economic conversion: 34,931 villages.

Pattern: High Road Connectivity + High Energy Access + Low Financial Inclusion + Low Agricultural Markets + Low Cottage and Traditional Industry.

Roads and energy are present, while financial, market, and enterprise systems remain weak.

The same logic can identify opportunities. High Livelihoods: Common Resources with Low Livestock and Veterinary, or High Livelihoods: Fisheries with Low Financial Inclusion and Low Agricultural Markets, are category-level filters that can be checked against their normalised feature values and then verified locally.

Spatial Analysis

Figure 2 shows the next step after choosing the Water and Sanitation category. It maps the village-level `water_sanitation_cat_value` across India as a smooth display surface. Each hexagon summarises nearby mapped village values for legibility at national scale; it is not a district or state average. The category class remains the filter, while this normalised value shows the strength and continuity of the spatial pattern.

Mission Antyodaya 2020: Village Water & Sanitation

Hexagonal display of 545,874 spatially linked village scores; each cell shows the mean of villages inside it.

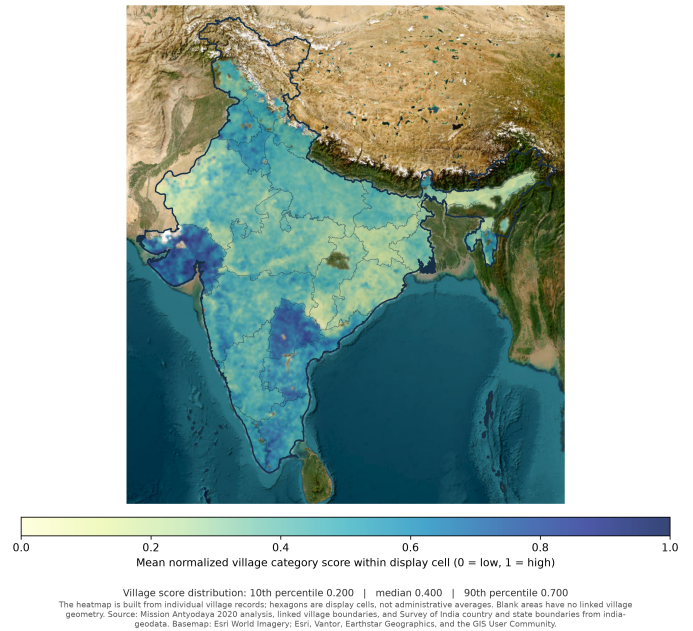


Figure 2: Village-derived Water and Sanitation category-value heatmap across India.

Users can return from the national surface to individual village category classes, compare the normalised category and feature values, check coverage, and finally trace the relevant raw survey entries. The same sequence works for all 21 categories.

Limitations

The classes are relative groupings in the 2020 data, not service standards, administrative rankings, causal findings, or programme recommendations. Missing values, uneven reporting, aggregation, ecology, and local economic history can affect interpretation. Results should be checked against feature definitions, more recent data, and field conditions. Raw survey fields can help trace a result, but cannot replace the normalised analytical values.

Conclusion

The value of the framework lies in moving between levels of evidence: from a category class, to its value, to the contributing normalised features, and finally to the raw survey entries that need checking. Used in this order, the analysis can narrow a field question without pretending that an unnormalised survey count is itself a comparable indicator.

References and data: [Mission Antyodaya 2020 Village Cluster Analysis Report](#); [village-level normalized values and cluster labels](#); [pan-India GeoPackage](#); [Mission Antyodaya 2020 raw data files](#); [GEE Antyodaya 2020 Explorer](#). Data preparation and analytical assistance: Ananya Singh, IIT Delhi.